北京郵電大學

Queen Mary University of London

For examiners' use only

1	
2	
3	
4	
Total	

EBU6304 A

Joint Programme Examinations 2022/23

EBU6304 Software Engineering

Paper A

Time allowed 2 hours

Answer ALL questions

Complete the information below about yourself very carefully.

QM student number										
BUPT student number										
Class number										

NOT allowed: electronic calculators and electronic dictionaries.

INSTRUCTIONS

1. **You must NOT take answer books, used or unused, from the examination room.**
2. Write only with a black or blue pen **and in English**.
3. Do all rough work in the answer book – **do not tear out any pages**.
4. If you use Supplementary Answer Books, tie them to the end of this book.
5. Write clearly and legibly.
6. **Read the instructions on the inside cover.**

Examiners

Dr Ling Ma, Dr Gokop Goteng, Dr Matthew Huntbach, Dr Ethan Lau

Copyright © Beijing University of Posts and Telecommunications & © Queen Mary University of London 2022

Instructions

Before the start of the examination

1) Place your BUPT and QM student cards on the corner of your desk so that your picture is visible.

2) Put all bags, coats and other belongings at the back/front of the room. All small items in your pockets, including wallets, mobile phones and other electronic devices must be **placed in your bag in advance**. **Possession of mobile phones, electronic devices and unauthorised materials is an offence.**

3) Please ensure your mobile phone is switched off and that no alarm will sound during the exam. **A mobile phone causing a disruption is also an assessment offence.**

4) Do not turn over your question paper or begin writing until told to do.

During the examination

1) You must not communicate with or copy from another student.

2) If you require any assistance or wish to leave the examination room for any reason, please raise your hand to attract the attention of the invigilator.

3) If you finish the examination early you may leave, but not in the first 30 minutes or the last 10 minutes.

4) For 2 hour examinations you may **not** leave temporarily.

5) For examinations longer than 2 hours you **may** leave temporarily but not in the first 2 hours or the last 30 minutes.

At the end of the examination

1) You must stop writing immediately – **if you continue writing after being told to stop, that is an assessment offence.**

2) Remain in your seat until you are told you may leave.

Question 1

a) List the advantages and potential problems of reusing existing software components in a new software product and discuss if it can save the cost of new software product development.

[6 marks]

	Do not write in this column
	6 marks

b) A university is planning to develop a virtual reality classroom system to enhance remote teaching and learning experience. Suggest the most appropriate *software process model* that might be used for developing this system. Give your reasons.

[4 marks]

	Do not write in this column
	4 marks

c) One of the non-functional requirements in a mobile APP development is "*energy-efficient: make the APP more usable on low power mobile devices and help reduce the overall carbon footprint*". Suggest a method to verify this requirement and re-write this non-functional requirement to make it verifiable.

[4 marks]

	Do not write in this column
	4 marks

d) You are going to test a software unit which is to validate a username format. A valid username must be six characters long, the first three characters must be letters and the last three characters must be digits. For example, *abc123* and *ghh558* are valid usernames. Design test cases using *Partition Testing*.

[11 marks]

	Do not write in this column

	11 marks

Question marking: $\frac{}{6} + \frac{}{4} + \frac{}{4} + \frac{}{11} = \frac{}{25}$

Question 2

a) Answer the questions below about *Analysis* and *Design*:

[5 marks]

i) "*Refactoring means frequently changing the code in order to change its external behaviour*". Do you agree with this statement? Explain why.

(3 marks)

ii) What is the *multiplicity* of an association? Give an example.

(2 marks)

	Do not write in this column
	5 marks

b) Answer the questions below about *Software Architecture*:

[20 marks]

i) It is often said that "*non-functional requirements are a result of the architectural choices*". Discuss an architectural choice in the context of web-based applications that makes the system more robust to hardware failures.

(3 marks)

ii) A web-based application your company is developing needs to be RESTful, i.e., that it respects the *REST architectural constraints*. In the current version, after a user has logged in, the client is

still required to send its login credentials to the server at every request. Which of the *REST constraints* does this behaviour follow or violate and why?

(6 marks)

iii) "*Architectural patterns can be seen as a way of reusing knowledge*". Do you agree with this statement? Explain why.

(4 marks)

iv) What is *load balancing* in the context of distributed systems and what is its aim?

(4 marks)

v) What is the *business layer* in the context of mobile applications architecture and how can it be used to reduce the computational load on the mobile?

(3 marks)

	Do not write in this column

	20 marks

Question marking: $\frac{}{5} + \frac{}{20} = \frac{}{25}$

Question 3

a) Answer the questions below about *Project Management*:

[10 marks]

i) What are the goals of a successful software *project management*? Explain.

(6 marks)

ii) In *project scheduling*, we should not assume that every stage will be problem-free. Suggest scheduling rules to deal with unexpected problems.

(4 marks)

	Do not write in this column
	10 marks

b) FinSoft is a Financial Technology company that provides applications for financial services. FinSoft has its headquarter in New York and opened new branches in London, Beijing, Dubai and South Africa last year. The company used physical servers in New York, but to save cost, the company decided to use Virtual Machines (VMs) running on Azure Cloud as virtual servers in all new branches. The company implemented the VMs at once on the same day as they launched the new cloud system. Unfortunately, the system crashed in all the new branches. The company did not have enough staff resources with the required cloud technology skills to troubleshoot cloud systems. As a result, they lose revenue and customers.

[10 marks]

i) Identify and discuss THREE project management issues that caused this failure.

(6 marks)

ii) Suggest TWO short-term solutions that the company can adopt to solve the problems at those new branches.

(2 marks)

iii) Suggest TWO long-term solutions that the company can adopt to solve the problem at those new branches.

(2 marks)

	Do not write in this column

	10 marks

c) Describe the features of *Open-Source Software* (OSS) and give one example of OSS.

[5 marks]

	Do not write in this column
	5 marks

Question marking: $\frac{}{10} + \frac{}{10} + \frac{}{5} = \frac{}{25}$

Question 4

a) In *Object-Oriented Programming*, objects are defined by classes with variables and methods. Objects interact with each other by calling a public method on an object of that class or by calling a public static method from the class. Sometimes the code has to be modified although it works correctly. For what reasons it may be a good idea to modify the code inside a class without changing the way other classes interact with it? Explain your reasons.

[10 marks]

	Do not write in this column

		10 marks

b) You may need to make a modified version of an original class, without changing the code of the original class. Objects of the modified version will work slightly differently when methods are called, but they can be passed as arguments to methods in the original class. One way of doing this is to use *inheritance*, that is to write a subclass of the original class. Another way is to use the *Decorator design pattern*. Explain how the *Decorator design pattern* works, how what it produces is different from a subclass, and how in some cases it is better to use it rather than *inheritance*.

[8 marks]

	Do not write in this column

		8 marks

c) Suppose we have a class `Account` which has a constructor with the header

```
Account(String name, int accountNumber)
```

where `name` gives the name of the owner of the `Account`, and `accountNumber` gives the number of the `Account`, with every `Account` meant to have a different number. There are also subclasses of `Account` called `SavingAccount` and `BorrowAccount` that take the same arguments in their constructor. The class `Account` includes methods with headers `double balance()` which returns the current balance of the `Account`, and `void deposit(double value)` which adds what `value` gives to the balance.

Write a class `AccountFactory` that creates an object to create `Accounts` by calling a method on it. The call on it to create and return a new `Account` has header:

```
Account createAccount(String name, double value, String type)
```

where `name` gives the owner of the `Account` and `value` gives its balance when it is returned. If type is set to `"Saving"`it returns a `SavingAccount`, and if it is set to `"Borrow"` it returns a `BorrowAccount`. The `AccountFactory` object decides what number to give for the `accountNumber` of an `Account`.

This is an example of the *Factory Object design pattern*.

[7 marks]

	Do not write in this column

Do not write in this column

2022-2023
Rough Working
Page 16 of 18

Do not write in this column

2022-2023
Rough Working
Page 17 of 18

Do not write in this column

2022-2023
Rough Working
Page 18 of 18